Welcome to MCS 481

1. ## About the Course
   - content and organization of the course
   - goals, expectations, and evaluation
   - CGAL: the Computational Geometry Algorithms Library

2. ## The Convex Hull Problem
   - definition of convexity
   - a first algorithm
   - cost analysis
   - degenerate cases

MCS 481 Lecture 1
Computational Geometry
Jan Verschelde, 24 August 2026

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What is computational geometry?

Consider the following problems:

1. Locate the nearest coffee shop.
   A Voronoi diagram is a partition of the plane into regions such that all points in the the same region have the same coffee shop as closest.
2. Get to the coffee shop avoiding obstacles (e.g. buildings).
   This is a robot motion planning problem using maps.
3. Overlay of maps, maps with shop locations, roads and buildings.

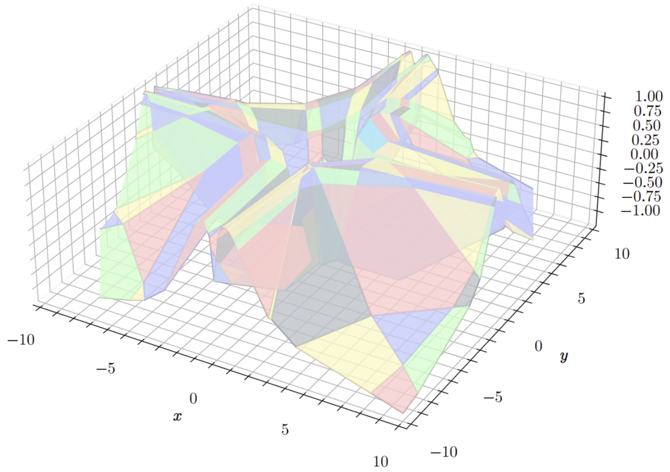
Definition (computational geometry)

Computational geometry studies algorithms and data structures to solve geometric problems with focus on exact computations that are asymptotically fast.

an application: linear regions in neural networks

joint with Johnny Joyce, CASC 2025

A linear region in a neural network is a connected region for which the map defined by the neural network is linear.



Content and Organization of the Course

Textbook: *Computational Geometry. Algorithms and Applications* by M. de Berg, M. van Kreveld, M. Overmars, and O. Schwartzkopf.

We will cover the first 11 chapters in depth (skim the last 5):

1. Computational Geometry, an introduction
2. Line Segment Intersection, thematic map overlay
3. Polygon Triangulation, guarding an art gallery
4. Linear Programming, manufacturing with molds
5. Orthogonal Range Searching, querying a database
6. Point Location, knowing where you are
7. Voronoi Diagrams, the post office problem
8. Arrangements and Duality, supersampling in ray tracing
9. Delaunay Triangulations, height interpolation
10. More Geometric Data Structures, windowing
11. Convex Hulls, mixing things

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Purpose of the Course

Prerequisite(s): Grade of C or better in MCS 401,
or MCS 471 *(not yet official)*; or consent of the instructor.

The goals of MCS 481 are

1. understand statement and properties of geometric problems;
2. know algorithmic techniques to solve geometric problems; and
3. apply software to explore geometric algorithms.

There will be three computer projects during the semester.
On projects, you are encouraged (not required) to work in pairs.

There are two exams:

1. Each midterm exam is worth 100 points.
2. The final exam counts for 200 points.

The final exam may be substituted by a final project.

Homework: 100 points, collected (roughly) every two weeks.

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The Computational Geometry Algorithms Library

http://www.cgal.org

MCS 481 is not a programming course.
Neither is this a course where all examples are computed by hand.
CGAL provides an extensive collection of examples.

We use CGAL, or other software for computational geometry,
to experiment with asymptotic cost and case analysis.

- CGAL is available via package managers.
- There is a Python interface, install with pip or conda.

Lecture 3 will be devoted to software for computational geometry.

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the convex hull of a set

In a convex set, a line segment between any two points in the set is entirely contained in the set. Formally:

Definition (convexity)

A set S is *convex* if and only if

$$\text{for all } x, y \in S, \text{for all } t \in [0,1] : (1-t)x + t\,y \in S.$$

Definition (convex hull of a set)

Given a set S, the *convex hull of S*, denoted by $\mathrm{CH}(S)$ is the intersection of all convex sets that contain S.

Imagine points as nails,
take a rubber band and let it snap around the points.

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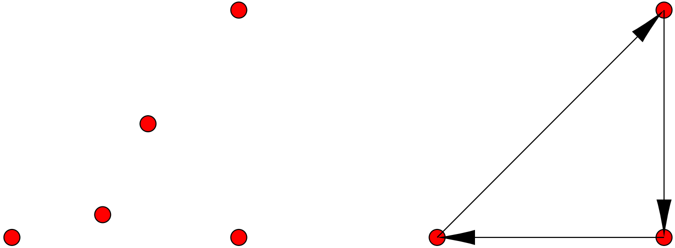
specification of input and output

Input: n points in $\mathbb{R}^2$, $S = \{p_0, p_1, \ldots, p_{n-1}\}$, $\#S = n < \infty$.
Output: list of points sorted in clockwise fashion
so two consecutive points in the list span an edge:
$L = [q_0, q_1, \ldots, q_{m-1}]$, $\left(q_i, q_{i+1 \bmod m}\right)$ spans an edge,
for $i = 0, 1, \ldots, m-1$.

$S = \{(4,1), (0,0), (10,0), (6,5), (10,10)\}$, $L = [(0,0), (10,10), (10,0)]$



redefining the convex hull of a set

Definition (convex hull as a list)

A list L of m points, $L = [p_0, p_1, \ldots, p_{m-1}]$, defines a *convex hull* if for any consecutive pair of points $\left(p_i, p_{i+1 \bmod m}\right)$, all points lie at the same side of the line through p_i and $p_{i+1 \bmod m}$.

Exercise 1: Show that the above definition is equivalent to the earlier definition, repeated below.

Definition (convex hull of a set)

Given a set S, the *convex hull of S*, denoted by $\mathrm{CH}(S)$ is the intersection of all convex sets that contain S.

the idea for a first algorithm

Apply the criterion for a pair of points to be an edge.

Given is a finite set S of points.

0. Denote by E the list of edges.
1. Take all pairs of points $(p, q) \in S \times S$.
2. If all points $r \in S \setminus \{p, q\}$ lie at the same side
   of the line spanned by p and q, then add (p, q) to E.
3. Sort E so edges appear in clockwise order.

Exercise 2: Relate the *at the same side* to the orientation of the edges, clockwise or counter clockwise.

pseudo code for the first algorithm

Input: S, a finite set of points in $\mathbb{R}^2$.
Output: $L = \mathrm{CH}(S)$, the convex hull of S.

0. $E := \emptyset$
1. for all $(p, q) \in S \times S$, $p \neq q$, do
2.   if for all $r \in S \setminus \{p, q\}$, r is at the right of (p, q),
3.   then $E := E \cup (p, q)$
4. construct the sorted L from E

skipped some steps?

How do we go about this line:

2. if for all $r \in S \setminus \{p, q\}$, r is at the right of (p, q),

Exercise 3: Define the subroutine for the above step, for any set S and for any two points $p \neq q$.

How about the last step:

4. construct the sorted L from E

Exercise 4: Explain how quicksort (or any other $O(n \log(n))$ sorting algorithm) applies to the last step in the algorithm.

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cost analysis of the first algorithm

Input: S, a finite set of points in $\mathbb{R}^2$.
Output: $L = \mathrm{CH}(S)$, the convex hull of S.

0. $E := \emptyset$
1. for all $(p, q) \in S \times S$, $p \neq q$, do
2. if for all $r \in S \setminus \{p, q\}$, r is at the right of (p, q),
3. then $E := E \cup (p, q)$
4. construct the sorted L from E

The dimension of the problem is the size of S, let $n = \#S$.

1. For all $(p, q) \in S \times S$, $p \neq q$: $n(n-1)$ pairs.
2. From Exercise 3, $\#(S \setminus \{p, q\}) = n - 2$ steps.
3. We can sort in time $O(n \log(n))$.

Multiplying $O(n^2)$ with $O(n)$ gives an $O(n^3)$ time algorithm.

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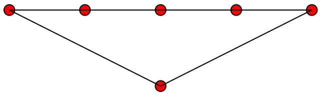
two degenerate cases

1. Lattice polytopes:



Origin: $p(x, y) = x^0y^1 + x^1y^1 + x^2y^1 + x^3y^1 + x^4y^1 + x^2y^0$.

Problem: same edge is reported multiple times.

2. Approximate coordinates:

Problem: report too many or too few edges.

recommended exercises

We covered pages 1 through 5 in the textbook.

Consider the following activities, listed below.

1. Browse the site `www.cgal.org` and explore.
2. Install JupyterLab (using miniconda) on your computer.
3. Install CGAL, at least the Python interface.
4. Write the solutions to Exercises 1 through 4.
5. Consider the exercises 1,2,3 in the textbook.

A selection of homework problems will be collected later.